

BSc (Hons) Computer Science
BSc (Hons) Cyber Security
Module Code: QH0305 Module Title: Problem Solving
Assessment Sheet 2

Instructions:

This is one of the eight assessment tasks which will contribute to the overall mark. You will need to complete the tasks as outlined below and then document them in a Word document file. As a minimum, you should provide screenshots of the following:

- Your code
- The output that your code generates

In instances where your code could generate different outputs depending on what values are given, you should provide multiple screenshots of the console screen, showing the different outputs in order to demonstrate that the code works correctly.

Your Word document should have appropriate headings to ensure that each task can easily be identified alongside the rest of your work.

This assessment will focus on switch/case structure.

You must attempt all tasks on this sheet to achieve a higher grade. For example, if you want to gain marks between 70 - 100, you must complete all other grades first and add them to your portfolio with screenshots.

A zip folder with all Grade codes (in .c or .txt format) must be submitted alongside the portfolio (MS Word file). This is in addition to the code presented as screenshots.

Assessment Task 2: City Aquarium Ticketing

Task Objective:

Create a program for an aquarium ticketing system that calculates the total cost of tickets based on user selections. The program must use switch/case for price determination (and for the menu in the top band)

To achieve Grade D (Between 40-49) (Basic Ticket Price Calculation, Fixed Prices)

1. Create variables to store:

- **zone** (exhibit zone: 'S' = Sharks, 'R' = Reefs, 'P' = Penguins)
- **ticketType** ('N' = Standard Entry, 'E' = Express Entry)
- **ticketPrice** (price of one ticket, derived from zone + ticketType)

Choose any fixed default values for **zone** and **ticketType** (no user input).

2. Use a **switch statement** to set ticket prices based on the following:

- Sharks: £22 (Standard), £32 (Express)
- Reefs: £18 (Standard), £28 (Express)
- Penguins: £16 (Standard), £26 (Express)

3. Display the zone, ticket type, and the ticket price (formatted to 2 d.p.).

Example Outcome:

Zone: Sharks

Ticket Type: Standard

Ticket Price: £22.00

To achieve Grade C (Between 50-59) (User Input and Multi-Ticker Total)

Build on 40–49; retain switch-based pricing.

1. Prompt the user to select:
 - Zone (S, R, P)
 - Ticket type (N, E)
 - Number of tickets (integer)
2. Validate that the number of tickets > 0; otherwise print an error and stop.
3. Use a switch-case structure on the selected zone (and ticket type where needed) to determine price per ticket. Assume that all tickets purchased together are for the same zone and of the same type.
4. Compute and display: zone, ticket type, quantity, total cost.

Example Outcome:

Zone: Reefs

Ticket Type: Express

Tickets: 3

Total Cost: £84.00

To achieve Grade B (Between 60-69) (Bulk Discount and Optional Add-On)

Build on 50–59; keep the same price table and switch logic.

1. Bulk discount: if tickets > 5, apply 15% discount to the pre-add-on total.
2. Optional add-on: ask if the user wants an Audio Guide (y/n) at £4.50 per ticket.
3. Calculate and display:
 - Zone, ticket type, number of tickets
 - Total before discount, discount amount (if any)
 - Add-on cost, and final total cost
4. Maintain input validation (tickets > 0; zone and ticketType in allowed sets).

Example Interaction:

Zone: Penguins
Ticket Type: Standard
Tickets: 6
Audio Guide: y
Total Before Discount: £96.00
Bulk Discount (15%): -£14.40
Add-on Cost: £27.00
Final Total: £108.60

To achieve Grade A (Between 70-100) (Menu-Driven System)

Build on 60–69; the menu itself must be implemented with switch and repeat until exit.

1. Add a menu system using a switch statement:
 - Option 1: Book tickets
 - Option 2: View ticket prices (show the table for Sharks/Reefs/Penguins with Standard/Express)
 - Option 3: Exit the program
2. Allow the user to interact with the menu repeatedly without restarting the program
3. Validate all inputs (invalid menu choice, zone, ticket type, ticket count).
4. After each iteration, display a detailed summary (zone, type, qty, before-discount total, discount, add-on, final total).

Example Interaction (abridged):

```
--- City Aquarium Ticketing ---
1. Book Tickets
2. View Ticket Prices
3. Exit
Enter choice: 1
...
Final Total: £...
Return to menu? y
```

Assignment Preparation Guidelines

- All components of the assignment report must be Word-processed (**handwritten text or hand drawn diagrams are not acceptable**), font size must be within the range of 11 point to 14 point including the headings, body text and any texts within diagrams.
- Standard and commonly used fonts such as Times New Roman, Arial or Calibri should be used.
- All figures, graphs and tables must be numbered and labelled with short explanations.
- Material from external sources must be properly acknowledged and cited within the text using the Harvard referencing system.
- All components of the assignment (text, diagrams, code etc.) and all assessment sheets must be submitted in one Word file.
- The report should be logically structured, the core of the report may start by defining the problem / requirements, followed by the proposed solution including a detailed discussion, analysis and evaluation, leading to the implementation and testing stage, finally a conclusion and personal reflection on learning.
- Screenshots without description / discussion are not suitable as they do not express your understanding or support your work adequately.

Submission instructions

- This is a portfolio assignment with eight tasks in total. Each task will be completed and saved in the portfolio. Once the portfolio is completed, it should be submitted on Turnitin. The submission link to Turnitin can be found under the “Assessment Tab” in your module section in the SOL VLE.
- Please note file size limitation might apply. Your report must be under 250MB.
- The source code (in .c or .txt format) for each task should be **zipped** and complete single ZIP file **submitted** to the VLE alongside the report.
- The Assignment Brief can be found under the “Assessment Tab” in your module section in the SOL VLE.
- **Refer to the Assignment Brief** to find the links to Late Submissions, Extenuating Circumstances, Academic Misconduct, Ethics Policy, Grade marking and Guidance for online submission through Solent Online Learning (SOL).